

PROJECT REPORT

Machine Learning for Cancer Diagnosis

Personal Professional Project

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1 Introduction

1.1 Global Challenge of Cancer

Cancer remains one of the most formidable challenges to global health in the 21st century. According to the World Health Organization (WHO), cancer is a leading cause of death worldwide, accounting for nearly 10 million deaths annually. With global cancer cases expected to rise by 60 percent over the next two decades, the pressure on healthcare systems to provide rapid and accurate diagnoses has never been greater.

1.2 The Problem : Complexity and Human Limitations

The diagnostic process involves analyzing a vast array of cellular characteristics. A single biopsy can produce dozens of measurements, such as cell radius, texture, perimeter, and concavity. For a human pathologist, consistently weighing over 30 different variables simultaneously is an immense cognitive task.

1.3 The Solution : Bridging the Gap with Machine Learning

Machine Learning acts as a bridge between raw biological data and clinical action. By using the Random Forest algorithm, we can process high-dimensional data instantly and objectively. This technology provides :

- **Mathematical Objectivity** : Removing subjective bias from the diagnostic flow.
- **Increased Sensitivity** : Identifying malignancy at the earliest possible stage.
- **Workflow Efficiency** : Allowing doctors to spend more time on patient care.

1.4 Aim of the Project

The primary objective is to build and evaluate a robust diagnostic support tool. Using a dataset of clinical cell measurements, the goal is to implement a Random Forest classifier that can accurately distinguish between malignant and benign tumors.

2 State of the Art

2.1 Machine Learning Definition

Machine learning is a technique that allows a computer to learn from data to make predictions or decisions. In the medical field, it is used to analyze patient data and help doctors make faster and more accurate decisions.

2.2 Techniques Used

Several machine learning techniques are used for cancer diagnosis :

1. **Logistic Regression** : A simple classification method
2. **K-Nearest Neighbors (KNN)** : Classifies data by comparing with similar cases
3. **Decision Trees and Random Forest** : Used to make decisions based on conditions
4. **Support Vector Machine (SVM)** : Separates data into different categories
5. **Neural Networks** : Complex models for advanced analysis

2.3 Applications

Machine learning is used to :

- Detect abnormalities in medical data
- Classify tumors as benign or malignant
- Enable early detection of cancer
- Support medical decision-making

2.4 Results in Literature

Breiman (2001) introduced the Random Forest algorithm, which combines multiple decision trees to improve prediction accuracy. In the literature, many studies show that machine learning models achieve good results in cancer diagnosis, with some models reaching accuracy above 90 percent. However, results depend on the dataset and data quality.

2.5 Limitations

Machine learning has limitations in cancer diagnosis :

- Requires large amounts of high-quality data
- Some models are difficult to interpret
- Can produce false positives or false negatives
- Ethical issues related to medical data

3 Data Preprocessing

3.1 Data Loading

The dataset was obtained from Kaggle and loaded into a Pandas DataFrame. Each row represents a patient, and each column corresponds to a clinical measurement. The target variable indicates whether the tumor is benign (B) or malignant (M).

3.2 Data Cleaning

3.2.1 Removal of Irrelevant Columns

The id column was removed as it carries no predictive information.

3.2.2 Handling Missing Values

The dataset was inspected for missing values. No significant missing entries were found.

3.2.3 Removing Duplicates

Duplicate rows were removed to prevent bias.

3.2.4 Encoding the Target Variable

The target column was encoded as :

- $M \rightarrow 1$ (malignant)
- $B \rightarrow 0$ (benign)

3.3 Feature-Target Separation

The dataset was split into :

- $\mathbf{X}$: all feature columns
- $\mathbf{y}$: the target variable

3.4 Train-Test Split

The dataset was divided into :

- 80% training data
- 20% testing data

Stratification preserved the proportion of cases in both sets.

3.5 Feature Scaling

A StandardScaler was used to standardize each feature :

$$x_{\text{scaled}} = \frac{x - \mu}{\sigma}$$

The scaler was fitted only on training data to avoid data leakage.

4 Training of the Data

4.1 Algorithm Selection : Random Forest

Random Forest Classifier was selected for its robustness and efficiency with medical datasets. Unlike single Decision Trees, Random Forest is an ensemble method that reduces variance and captures complex relationships between features.

4.2 Why 100 Estimators ?

The number of estimators was set to 100. Research shows that 100 trees provide optimal balance between accuracy and computational cost.

4.3 Evaluation Metrics

In cancer diagnosis, simple accuracy is insufficient. We prioritize :

4.3.1 Recall (Sensitivity)

Recall measures the model's ability to identify all positive cases. High Recall ensures the model catches as many actual cases as possible.

4.3.2 Precision

Precision ensures that when the model identifies a tumor as malignant, it is accurate.

4.3.3 The F1-Score

The F1-Score balances Precision and Recall :

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

5 Evaluation of the Model

The model achieved a global accuracy of 96.49% on the test set, with only 4 misclassifications out of 114 samples.

5.1 Results Summary

- **Global Accuracy** : 96.49%
- **Misclassifications** : 4 out of 114 samples
- **F1-Score (Benign)** : 0.97
- **F1-Score (Malignant)** : 0.95

- **Clinical Safety** : High
- **Diagnostic Precision** : High

6 Conclusion

A diagnostic support tool was successfully developed using the Random Forest algorithm to classify breast cancer tumors. The model achieved an overall accuracy of 96%, aligning with the goal of minimizing false negatives in cancer diagnosis.

Through feature importance analysis, characteristics such as concave points and worst area were identified as the most critical predictors. While this algorithm is not intended to replace human doctors, it serves as a powerful second opinion to reduce human error.

This project confirms that integrating Machine Learning into medicine is a vital step toward more accurate, data-driven healthcare.